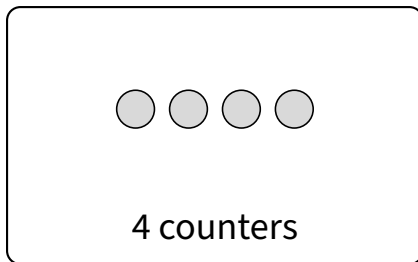
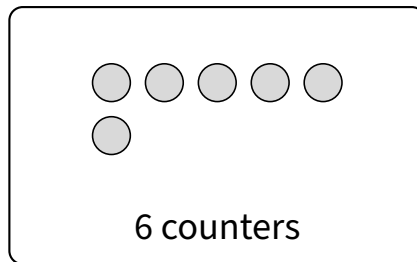


Each game is for two players and a pile of counters. The players take turns taking counters from the pile. Whoever takes the last counter wins.

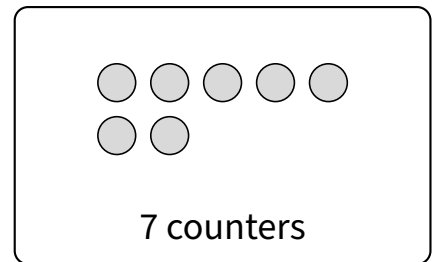
Problem 1: In this game, each turn takes 1 or 2 counters. Play with each starting pile below until you are sure whether you would rather go first or second, and circle your choice.



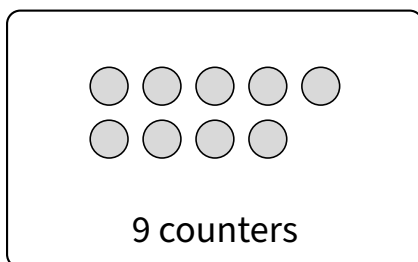
first second



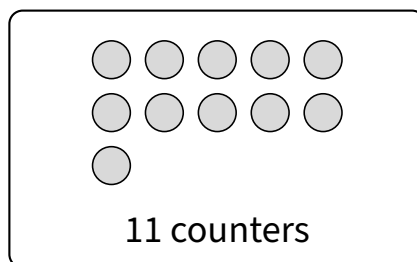
first second



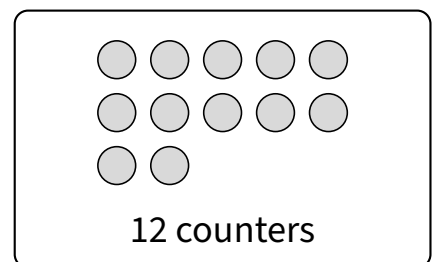
first second



first second



first second



first second

Problem 2: Each child below has a plan for the game where a turn takes 1 or 2 counters. Decide whether each plan wins every time, whatever the partner does, and circle your answer. If a plan can lose, write down a game where it loses by listing how many counters are left after each turn.

Mia: *The pile has 8 counters and I go first. I take 2. After that, when my partner takes 1, I take 2, and when my partner takes 2, I take 1.*

wins every time
can lose

Leo: *The pile has 10 counters and I go first. I always take 2, unless only 1 counter is left.*

wins every time
can lose

Ana: *The pile has 9 counters and I go second. When my partner takes 1, I take 2, and when my partner takes 2, I take 1.*

wins every time
can lose

Sam: *The pile has 11 counters and I go first. I take 1. After that, when my partner takes 1, I take 2, and when my partner takes 2, I take 1.*

wins every time
can lose

Problem 3: Now each turn takes 1, 3, or 4 counters. For every starting pile from 1 to 20 counters, decide whether you would rather go first or second. Write 1st or 2nd under the number.

1	2	3	4	5	6	7	8	9	10

11	12	13	14	15	16	17	18	19	20

Problem 4: Each turn still takes 1, 3, or 4 counters. For starting piles of 25, 30, 37, and 50 counters, would you rather go first or second? Explain how you know.

25 counters first second

30 counters first second

37 counters first second

50 counters first second

Problem 5: Each turn takes 1, 3, or 4 counters. Decide whether each plan below wins every time, whatever the partner does, and circle your answer. If a plan can lose, write down a game where it loses. If a plan wins every time, explain why it can never lose.

Kai: *The pile has 10 counters and I go first. On every turn, I take the largest number I am allowed to.*

wins every time
can lose

Theo: *The pile has 7 counters and I go second. On every turn, I take the same number my partner just took. If I can't, I take 1.*

wins every time
can lose

Zoe: *The pile has 9 counters and I go second. After each of my partner's turns, I leave 7 counters or 2 counters if I can. If I can't, I take all the counters that are left.*

wins every time
can lose

Problem 6: Jon says that when a turn may take 1, 2, or 4 counters, the starting piles where you would rather go second are different from when a turn may take only 1 or 2. For each rule, shade every starting pile from 1 to 15 where you would rather go second. Is Jon right? Explain how you can be sure, even for piles bigger than 15.

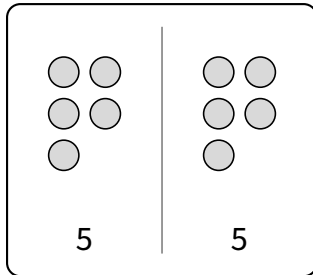
Take 1 or 2.

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
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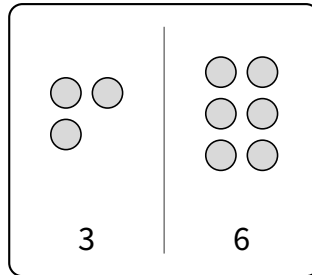
Take 1, 2, or 4.

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
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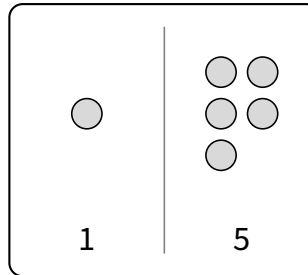
Problem 7: Now there are two piles. Each turn takes 1 or 2 counters from one of the piles, and whoever takes the very last counter wins. For each pair of piles below, circle whether you would rather go first or second. For the two piles of 5, explain why your choice wins every time.



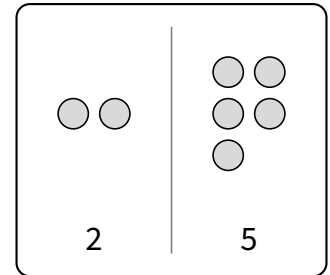
first second



first second



first second



first second